

Name: [REDACTED]

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Total
20.5
/22

Knowledge

1. Determine the derivative of each of the following. Express in fully factored form.

a) $f(x) = (e^{2x})(4 - e^{x^2-3})$
 $f'(x) = 2e^{2x}(4 - e^{x^2-3}) + e^{2x}(-2xe^{x^2-3})$
 $= e^{2x}(8 - 2e^{x^2-3} - 2e^{x^2-3})$
 $= e^{2x}(8 - 4e^{x^2-3})$
 $= 4e^{2x}(2 - e^{x^2-3})$

b) $g(x) = \frac{(3)^{x^2-8}}{x}$
 $g'(x) = \frac{2x(\ln 3)(3^{x^2-8})(x) - 3^{x^2-8}}{x^2}$
 $= \frac{3^{x^2-8}(2x^2 \ln 3 - 1)}{x^2}$

1.5 (2)

(2)

2. Determine the absolute minimum value of the function $f(x) = (5)^{1-x^2} + 4x^2$ is on the interval $-0.5 \leq x \leq 2$.

$f'(x) = -2x(\ln 5)(5^{1-x^2}) + 8x$
 $0 = -2x(\ln 5)(5^{1-x^2}) + 8x$
 $\downarrow$
 $x = 0$ no solution *It was 2!*

$f(-0.5) = 4.34 \rightarrow \text{Abs min}$
 $f(0) = 5$
 $f(2) = 16.008$

3 (4)

$\therefore$ Absolute minimum at $(-0.5, 4.34)$

Communication

1. Explain why the equation to the tangent of $g(x) = 2xe^{1-2x} + 1$, is a constant function at the point $A(0.5, 2)$.

Show your work mathematically.

$g'(x) = 2e^{1-2x} + 2x(-2)e^{1-2x}$
 $= e^{1-2x}(2 - 4x)$
 $0 = -2e^{1-2x}(2x - 1)$

$\therefore$ because the tangent is horizontal the equation must be a constant function

(3)

solution DNE $\downarrow$
 $x = \frac{1}{2} = 0.5$
 There is a horizontal tangent at 0.5

$g'(0.5) = -2e^0(1-1)$
 $= -2(0) = 0$
 tangent at $(0.5, 2)$
 $y = 0x + b \rightarrow y = 2$
 $2 = b$

9.5

2. Amy Stake is trying to find the critical values of $g(x)$. Is there anything wrong with her work? Explain.

$$g(x) = (2)^{2x^2-7x-4}$$

$$g'(x) = (2x^2 - 7x - 4)(2)^{2x^2-7x-4-1}$$

$$0 = (2x + 1)(x - 4)(2)^{2x^2-7x-5}$$

$$\therefore x = \frac{1}{2}, x = 4$$

$$0 = 2x + 1$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

she also found the zeros wrong

of $g'(x)$

Amy multiplied $g(x)$ by the exponent, rather than the derivative of the exponent. She also subtracted 1 from the exponent which she shouldn't have done. She also forgot to multiply by $\ln 2$.

$$g'(x) = (4x - 7)(\ln 2)(2)^{2x^2-7x-4}$$

$\downarrow$ instead
 $\downarrow$ answer
 $x = \frac{7}{4}$

(2)

Application

1. The value of a computer decreases in value as time goes on. Its value is modeled by the function, $V(t) = 2750(0.78)^t$, where V , is the value in dollars and t is the time from purchase in years.

a) What is the initial value of the computer?

$$V(0) = 2750(1) = \$2750$$

computer is worth \$2750 at the start

(1)

b) What is the rate of change for the value of the computer at $t = 15$?

$$V'(t) = 2750(\ln 0.78)(0.78)^t$$

$$V'(15) = 2750(\ln 0.78)(0.78)^{15} = -16.44$$

The computer is losing \$16.44 worth of value per year at 15 years

(2)

2. A small lake has a large population of bass. The bass population will grow according to the equation,

$$P(x) = \frac{1000}{-1+9e^{-x}}, \text{ where } P \text{ is the population of bass and } x \text{ is the time in years.}$$

a) Prove that the rate of change in the population of bass for any time t is $P'(x) = \frac{9000(e^{-x})}{(-1+9e^{-x})^2}$.

$$P'(x) = \frac{0(-1+9e^{-x}) - 1000(-9e^{-x})}{(9e^{-x}-1)^2} = \frac{9000e^{-x}}{(-1+9e^{-x})^2}$$

(2)

b) When does the rate of change of the bass population reach its maximum over the first four years?

$$0 \leq x \leq 4$$

$$P'(x) = \frac{9000e^{-x}}{(9e^{-x}-1)^2}$$

(4)

$$P''(x) = \frac{-9000e^{-x}(9e^{-x}-1)^2 - 2(9e^{-x}-1)(-9e^{-x})(9000e^{-x})}{(9e^{-x}-1)^4}$$

$$= \frac{(-9000e^{-x})(9e^{-x}-1)[9e^{-x}-1+18e^{-x}]}{(9e^{-x}-1)^4}$$

$$= \frac{(-9000e^{-x})(-9e^{-x}-1)}{(9e^{-x}-1)^3} = \frac{9000e^{-x}(9e^{-x}+1)}{(9e^{-x}-1)^3}$$

$$0 = 9000e^{-x}(9e^{-x}+1)$$

no solution

$$9e^{-x} = -1$$

$$e^{-x} = -\frac{1}{9}$$

no solution

$$P(0) = \frac{9000}{(9-1)^2} = \frac{9000}{64} = 140.63 \text{ bass/year}$$

$$P'(4) = \frac{164.94}{(-0.94)^2} = 236.33 \text{ bass/year}$$

∴ The max rate of change of bass per year for the first 4 years is 236.33 bass/year at the year 4 mark

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